

Chronic Kidney Disease Prediction Using Machine Learning Techniques

Phase 1 Presentation

Ömer Erkam İbiş

İzmir Institute of Technology

May 6, 2026

- **Chronic Kidney Disease (CKD):** A life-threatening condition defined as a progressive decline in kidney function over months or years .
- It is often referred to as a "silent killer" because it can cause kidney failure without exhibiting early symptoms .
- **Global Impact:** CKD affects approximately 10% of the global population, leading to millions of deaths due to inadequate treatment .
- **Research Objective:** To develop and validate predictive models for CKD using machine learning, addressing the complexity of large healthcare data .

The Role of eGFR in Diagnosis

- Early diagnosis is crucial to avoid severe outcomes like the need for dialysis or kidney transplants .
- The **Estimated Glomerular Filtration Rate (eGFR)** is the most accurate test used to measure kidney function and define the degree of the disease .
- eGFR is calculated using a combination of factors including blood creatinine levels, age, gender, diet, and albuminuria .
- The proposed predictive models aim to assess patient risk and calculate eGFR using the CKD_EPI equation to plan appropriate future treatments .

Related Work (Literature Review)

Previous studies have demonstrated the efficacy of machine learning in medical diagnostics:

- **Bemando et al. (2021):** Used Naïve Bayes and Random Forest for blood-related diseases, with Naïve Bayes showing high accuracy .
- **Kumar & Polepaka (2020):** Applied Random Forest and CNNs for medical disease prediction, where Random Forest outperformed others .
- **Sing et al. (2019):** Developed an SVM classifier for medical illness datasets, achieving up to 91% accuracy .
- **Desai et al. (2019):** Showed that Logistic Regression yielded highly accurate models compared to back-propagation neural networks for heart disease .

Dataset Description

- **Source:** Electronic medical records collected from the UCI Machine Learning Repository .
- **Instances:** The dataset contains 400 total instances .
 - 250 cases are classified as CKD .
 - 150 cases are classified as NOT_CKD .
- **Target Variable:** Binary classification where "1" denotes normal instances and "0" denotes sickness .

Feature Breakdown

The dataset consists of 25 total attributes used for training the classifiers .

11 Non-Categorical (Numerical) Attributes :

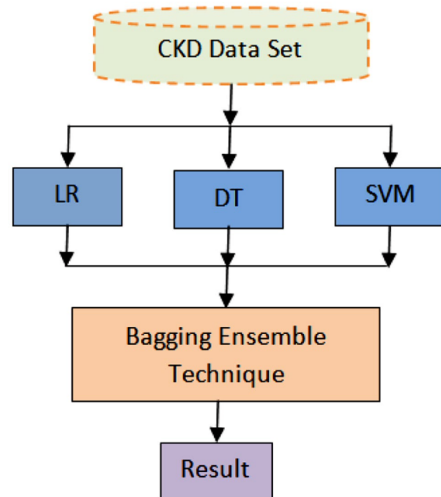
- Age, Blood Pressure (bp), Blood Glucose Random (bgr), Blood Urea (bu), Serum Creatinine (sc), Sodium (sod), Potassium (pot), Hemoglobin (hemo), Packed Cell Volume (pcv), White Blood Cell Count (wc), Red Blood Cell Count (rc) .

14 Categorical Attributes :

- Specific Gravity (sg), Albumin (al), Sugar (su), Red Blood Cells (rbc), Pus Cell (pc), Pus Cell Clumps (pcc), Bacteria (ba), Hypertension (htn), Diabetes Mellitus (dm), Coronary Artery Disease (cad), Appetite (appet), Pedal Edema (pe), Anemia (ane), and the target classification .

Proposed Methodology Architecture

- **Data Split:** The dataset was divided into 80% for training and 20% for testing.
- **Validation:** A 10-fold cross-validation method was applied during the training phase.
- **Stage 1 (Base Models):** The data was independently fed into three machine learning classifiers:
 - Logistic Regression (LR)
 - Decision Tree (DT)
 - Support Vector Machine (SVM)
- **Stage 2 (Ensemble Method):** A Bagging Ensemble Technique was subsequently applied to the base learners to reduce variance and improve overall performance .



Machine Learning Classifiers: Logistic Regression

- **Concept:** A statistical model that predicts the probability of an outcome in two values (0 to 1), unlike linear regression which predicts outside this range .
- Based on the logistic function ("logit"), it assumes independence of irrelevant alternatives .
- **Mathematical Representation:**

$$p = \frac{1}{1 + e^{-(a + b_1x_1 + b_2x_2 + \dots + b_px_p)}}$$

Machine Learning Classifiers: Decision Tree

- **Concept:** A supervised predictive modeling tool that generates a tree from the root using a top-down data partitioning approach .
- It utilizes multivariate analysis to split datasets based on multiple conditions and influences .
- It calculates splits using metrics like the Gini index and Entropy :

$$Gini = 1 - \sum_{i=1}^c (p_i)^2$$

$$Entropy = \sum_{i=1}^c -p_i \log_2(p_i)$$

Machine Learning Classifiers: Support Vector Machine (SVM)

- **Concept:** A supervised binary classification algorithm that separates data points to find an optimal hyperplane .
- Extremely efficient in handling outliers and operating in high-dimensional spaces .
- Uses decision functions known as *support vectors* .
- Can utilize various kernel types, including Linear, RBF, and Polynomial kernels .

Ensemble Technique: Bagging

- **Concept:** Also known as "Bootstrap Aggregation," it creates an ensemble of different base learners (like decision trees) .
- **Process:** Replicas of the input data are drawn independently with replacement. Each subset trains its own base learner .
- **Goal:** Primarily used to reduce the variance of a base learner .
- By averaging the predictions from a collection of different models, the final output becomes significantly more reliable than relying on a single base model .

Evaluation Metrics: Confusion Matrix

Performance is described using a Confusion Matrix, comparing true values against predictions :

- **True Positives (TP):** Positive instance correctly predicted as positive .
- **True Negatives (TN):** Negative instance correctly predicted as negative .
- **False Positives (FP):** Negative instance incorrectly predicted as positive (Type I Error) .
- **False Negatives (FN):** Positive instance incorrectly predicted as negative (Type II Error) .

Evaluation Metrics: Mathematical Formulas

Based on the Confusion Matrix, the following classification metrics were calculated :

- **Accuracy:** $ACC = \frac{TP+TN}{TP+FP+TN+FN}$
- **Recall (Sensitivity):** $Recall = \frac{TP}{TP+FN}$
- **Precision:** $Precision = \frac{TP}{TP+FP}$
- **Specificity:** $Specificity = \frac{TN}{TN+FP}$
- **F1-score:** $F = 2 \cdot \frac{Precision \cdot Recall}{Precision+Recall}$

Results: Base Classifiers Performance

Classifier	Precision	Recall	F1	Acc (%)
Logistic Reg.	0.96	0.96	0.96	93.28
SVM	0.96	0.96	0.96	94.80
Decision Tree	0.99	0.98	0.98	95.92

Table: Classification report

- The Decision Tree classifier performed the best among the independent base models .
- The boxplot illustrates the accuracy distributions, highlighting DT's superior performance and higher median accuracy compared to LR and SVM .

Table 4 Accuracy achieved by different Ensemble techniques

Ensemble Methods	Accuracy (%)		
	LR	DT	SVM
Bagging	94.53	97.23	95.70

Figure: Performance of base classifiers

Results: Bagging Ensemble Improvements

Ensemble Method	Base Classifier	Accuracy (%)
Bagging	Logistic Regression	94.53
Bagging	Support Vector Machine	95.70
Bagging	Decision Tree	97.23

Table: Accuracy achieved by Ensemble techniques

- The bagging method successfully increased the accuracy for all classifiers .
- The combination of **Decision Tree + Bagging** yielded the highest overall prediction accuracy at **97.23%** .

Comparison with Existing Literature

The proposed Decision Tree model outperforms several previously established frameworks :

Model	Accuracy (%)	Precision	Recall	F-score
Krishnamurthy (2020)	78.20	0.46	0.48	0.47
Rehman (2020)	88.30	0.66	0.86	0.74
Han (2019)	87.79	0.10	0.87	-
Proposed DT Model	95.92	0.99	0.98	0.98

Table: Comparison with other models (Excerpt from Table 5)

Conclusion & Future Scope

- **Conclusion:** Using the UCI chronic kidney disease dataset, the study validated that machine learning algorithms can highly accurately predict CKD .
- The Decision Tree paired with a Bagging Ensemble proved to be the most robust architecture, achieving a 97.23% accuracy rate .
- **Clinical Impact:** These predictive models serve as vital tools for medical practitioners to diagnose CKD early, recommend lifestyle changes, and ultimately save lives .
- **Future Work:** Model performance could be further optimized by applying specific feature selection methods to refine the dataset .

Thank You!